

# The Mathematics of Backpropagation: Companion Derivations for `llm.moj`

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**Abstract**—Working derivations of the backward pass for every operation in the GPT-2 forward pass, as implemented by the `llm.moj` kernels. Sections are ordered by backward-derivation pedagogy—starting from the scalar loss and working toward the inputs—rather than by `llm.c` forward order, so that each gradient is derived only after the one it depends on. Each section anchors on the forward computation and leaves the derivation as structured workspace.

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## I. NOTATION AND CONVENTIONS

Shapes follow the kernel code, as collected in Table I.

Indices:  $b$  over batch,  $t$  over positions,  $i, j, k$  over channels or vocab entries. A row of activations at one position is  $x_{bt} \in \mathbb{R}^C$ .

a) *The backward convention.*: Every op receives the upstream gradient  $\frac{\partial L}{\partial y}$  of the scalar loss  $L$  with respect to its output  $y$ , and must produce  $\frac{\partial L}{\partial x}$  for each input  $x$  (and  $\frac{\partial L}{\partial w}$  for each parameter) via the chain rule

$$\frac{\partial L}{\partial x_i} = \sum_j \frac{\partial L}{\partial y_j} \frac{\partial y_j}{\partial x_i}. \quad (1)$$

Gradients into shared inputs *accumulate* ( $+=$ ) because a tensor used by several consumers receives a contribution from each.

b) *Reading order.*: The forward pass runs encoder  $\rightarrow$  layernorm  $\rightarrow$  attention  $\rightarrow$  matmul  $\rightarrow$  GELU  $\rightarrow$  softmax  $\rightarrow$  cross-entropy. The sections below instead follow the order in which gradients are *derived*: starting at the loss (§III) and moving back toward the embeddings (§IX), each op reusing the upstream gradient established by the section before it. §X then reassembles these pieces into the end-to-end backward pass.

**TODO:** Decide and record: numerator vs. denominator layout, and where transposes land in the matmul gradients as a consequence.

## II. WARM-UP: SCALAR CHAIN RULE ON A TINY NETWORK

A two-step composition to fix conventions before the tensor ops:  $z = wx + b$ ,  $a = \sigma(z)$ ,  $L = \frac{1}{2}(a - y)^2$ .

**Derivation.** Backward through the loss, then the sigmoid:

$$\frac{\partial L}{\partial a} = a - y \quad (2)$$

$$\frac{\partial L}{\partial z} = \frac{\partial L}{\partial a} \cdot \frac{\partial a}{\partial z} \quad (3)$$

$$\frac{\partial a}{\partial z} = \sigma(z) \cdot (1 - \sigma(z)) \quad (4)$$

and since  $a = \sigma(z)$ ,

$$\frac{\partial L}{\partial z} = (a - y) \cdot \frac{\partial a}{\partial z} = (a - y) \cdot a \cdot (1 - a). \quad (5)$$

For the weight:

$$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial a} \cdot \frac{\partial a}{\partial z} \cdot \frac{\partial z}{\partial w} = \frac{\partial L}{\partial z} \cdot \frac{\partial z}{\partial w} \quad (6)$$

$$\frac{\partial z}{\partial w} = x \quad (7)$$

$$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial z} \cdot x = (a - y) \cdot a \cdot (1 - a) \cdot x. \quad (8)$$

For the bias:

$$\frac{\partial L}{\partial b} = \frac{\partial L}{\partial a} \cdot \frac{\partial a}{\partial z} \cdot \frac{\partial z}{\partial b} = \frac{\partial L}{\partial z} \cdot \frac{\partial z}{\partial b} \quad (9)$$

$$\frac{\partial z}{\partial b} = 1 \quad (10)$$

$$\frac{\partial L}{\partial b} = \frac{\partial L}{\partial z} = (a - y) \cdot a \cdot (1 - a). \quad (11)$$

TABLE I  
SYMBOLS AND THEIR CORRESPONDING KERNEL PARAMETERS.

| Symbol | Code              | Meaning                              |
|--------|-------------------|--------------------------------------|
| $B$    | batch_size        | batch size                           |
| $T$    | seq_len           | sequence length                      |
| $C$    | channels          | model width (embedding dim)          |
| $V$    | vocab_size        | real vocabulary size                 |
| $V_p$  | vocab_size_padded | padded vocabulary (row stride)       |
| $H$    | num_heads         | attention heads, head size $h = C/H$ |

For the input:

$$\frac{\partial L}{\partial x} = \frac{\partial L}{\partial a} \cdot \frac{\partial a}{\partial z} \cdot \frac{\partial z}{\partial x} = \frac{\partial L}{\partial z} \cdot \frac{\partial z}{\partial x} \quad (12)$$

$$\frac{\partial z}{\partial x} = w \quad (13)$$

$$\frac{\partial L}{\partial x} = \frac{\partial L}{\partial z} \cdot w = (a - y) \cdot a \cdot (1 - a) \cdot w. \quad (14)$$

**Result.** One chain back through the network, fanning out at the affine inputs:

$$\frac{\partial L}{\partial a} \rightarrow \frac{\partial L}{\partial z} \rightarrow \left\{ \frac{\partial L}{\partial w}, \frac{\partial L}{\partial b}, \frac{\partial L}{\partial x} \right\} \quad (15)$$

with  $\frac{\partial L}{\partial z} = (a - y) \cdot a \cdot (1 - a)$  computed once and reused:  
 $\frac{\partial L}{\partial w} = \frac{\partial L}{\partial z} \cdot x$ ,  $\frac{\partial L}{\partial b} = \frac{\partial L}{\partial z}$ ,  $\frac{\partial L}{\partial x} = \frac{\partial L}{\partial z} \cdot w$ .

### III. CROSS-ENTROPY LOSS

Standard cross-entropy between a target distribution  $y_{bt} \in \mathbb{R}^V$  and predicted probabilities  $p_{bt} \in \mathbb{R}^{V_p}$ :

$$L_{bt} = - \sum_{i < V} y_{bt,i} \log p_{bt,i}. \quad (16)$$

For language modeling the target distribution is one-hot. Writing  $\mathbb{K}[P]$  for the indicator of a condition  $P$ ,

$$\mathbb{K}[P] = \begin{cases} 1 & P \text{ is true,} \\ 0 & P \text{ is false,} \end{cases} \quad (17)$$

the target with token  $y_{bt} \in \{0, \dots, V-1\}$  is  $y_{bt,i} = \mathbb{K}[i = y_{bt}]$ : all probability mass on the target token, zero elsewhere. Every term of (16) but one then vanishes, and the loss collapses to the form computed by `crossentropy_oh_fwd`:

$$L_{bt} = - \log p_{bt, y_{bt}}. \quad (18)$$

#### A. Backward w.r.t. the probabilities

**Derivation.** Break the standard form (16) out term by term:

$$L_{bt} = - (y_{bt,0} \log p_{bt,0} + y_{bt,1} \log p_{bt,1} + y_{bt,2} \log p_{bt,2} + \dots + y_{bt,V-1} \log p_{bt,V-1}). \quad (19)$$

Differentiating with respect to one coordinate  $p_{bt,i}$ , every term of the broken-out sum except the  $i$ -th is constant, so only its derivative survives:

$$\begin{aligned} \frac{\partial L}{\partial p_{bt,i}} &= - \frac{\partial}{\partial p_{bt,i}} (y_{bt,0} \log p_{bt,0} + \dots + y_{bt,i} \log p_{bt,i} \\ &\quad + \dots + y_{bt,V-1} \log p_{bt,V-1}) \\ &= - y_{bt,i} \frac{\partial}{\partial p_{bt,i}} \log p_{bt,i}. \end{aligned} \quad (20)$$

The surviving factor is the derivative of the logarithm,

$$\frac{d}{dx} \log x = \frac{1}{x} \implies \frac{\partial}{\partial p_{bt,i}} \log p_{bt,i} = \frac{1}{p_{bt,i}}, \quad (21)$$

giving the standard-sum gradient, valid for *any* target distribution  $y_{bt}$ :

$$\frac{\partial L}{\partial p_{bt,i}} = - \frac{\partial}{\partial p_{bt,i}} \sum_{j < V} y_{bt,j} \log p_{bt,j} = - \frac{y_{bt,i}}{p_{bt,i}}. \quad (22)$$

Substituting the one-hot target  $y_{bt,i} = \mathbb{K}[i = y_{bt}]$ , (18) depends only on the target coordinate  $p_{bt, y_{bt}}$ :

$$\frac{\partial L}{\partial p_{bt,i}} = \begin{cases} -\frac{1}{p_{bt, y_{bt}}} & i = y_{bt}, \\ 0 & \text{otherwise.} \end{cases} \quad (23)$$

**Result.**

$$\frac{\partial L}{\partial p_{bt,i}} = - \frac{\mathbb{K}[i = y_{bt}]}{p_{bt,i}}. \quad (24)$$

### IV. SOFTMAX

Forward: from logits  $\ell_{bt} \in \mathbb{R}^{V_p}$  (real entries  $i < V$ ),

$$p_{bt,i} = \frac{e^{\ell_{bt,i} - m_{bt}}}{\sum_{j < V} e^{\ell_{bt,j} - m_{bt}}}, \quad m_{bt} = \max_{j < V} \ell_{bt,j}. \quad (25)$$

#### A. The Jacobian

Row indices  $bt$  are dropped throughout: each row is an independent softmax, so the Jacobian is per-row.

**Derivation.** The object here is the Jacobian entry  $\frac{\partial p_i}{\partial \ell_j}$  alone: how one probability moves with one logit. The loss enters only in the subsections that follow, where an upstream gradient is chained through these entries. Before differentiating: the max-subtraction in (25) cancels, because for any constant  $c$

$$\frac{e^{\ell_i - c}}{\sum_{j < V} e^{\ell_j - c}} = \frac{e^{-c} e^{\ell_i}}{e^{-c} \sum_{j < V} e^{\ell_j}} = \frac{e^{\ell_i}}{\sum_{j < V} e^{\ell_j}}, \quad (26)$$

so  $m_{bt}$  does not change the derivative and the unshifted form can be differentiated. Break the denominator sum out in the same format:

$$p_i = \frac{e^{\ell_i}}{S}, \quad S = e^{\ell_0} + e^{\ell_1} + e^{\ell_2} + \dots + e^{\ell_{V-1}}. \quad (27)$$

Every entry comes from the quotient rule: for  $u/v$  and a scalar argument  $x$ ,

$$\frac{\partial}{\partial x} \frac{u}{v} = \frac{\frac{\partial u}{\partial x} v - u \frac{\partial v}{\partial x}}{v^2}. \quad (28)$$

Work the  $2 \times 2$  block of entries in  $p_1, p_2$  and  $\ell_1, \ell_2$  explicitly before generalizing.

a) Entry  $\frac{\partial p_1}{\partial \ell_1}$ : Here  $u = e^{\ell_1}$  and  $v = S$ , and *both* move with  $\ell_1$ :

$$\frac{\partial u}{\partial \ell_1} = \frac{\partial}{\partial \ell_1} e^{\ell_1} = e^{\ell_1}, \quad (29)$$

$$\frac{\partial v}{\partial \ell_1} = \frac{\partial}{\partial \ell_1} (e^{\ell_0} + e^{\ell_1} + e^{\ell_2} + \dots + e^{\ell_{V-1}}) = e^{\ell_1}, \quad (30)$$

every term of the broken-out sum except  $e^{\ell_1}$  being constant. Substituting (29) and (30) into the quotient rule (28) with the values filled in:

$$\begin{aligned} \frac{\partial p_1}{\partial \ell_1} &= \frac{e^{\ell_1} (e^{\ell_0} + e^{\ell_1} + \dots + e^{\ell_{V-1}}) - e^{\ell_1} e^{\ell_1}}{(e^{\ell_0} + e^{\ell_1} + \dots + e^{\ell_{V-1}})^2} \\ &= \frac{e^{\ell_1} S - e^{\ell_1} e^{\ell_1}}{S^2} = \frac{e^{\ell_1}}{S} - \left( \frac{e^{\ell_1}}{S} \right)^2 \\ &= p_1 - p_1^2 = p_1 (1 - p_1). \end{aligned} \quad (31)$$

b) Entry  $\frac{\partial p_1}{\partial \ell_2}$ : Now the numerator does *not* move:  $u = e^{\ell_1}$  is constant in  $\ell_2$ , so  $\frac{\partial u}{\partial \ell_2} = 0$ , while again exactly one term of the broken-out denominator survives:

$$\frac{\partial v}{\partial \ell_2} = \frac{\partial}{\partial \ell_2} (e^{\ell_0} + e^{\ell_1} + e^{\ell_2} + \dots + e^{\ell_{V-1}}) = e^{\ell_2}. \quad (32)$$

Substituting into (28) with the values filled in:

$$\begin{aligned} \frac{\partial p_1}{\partial \ell_2} &= \frac{0 \cdot (e^{\ell_0} + e^{\ell_1} + \dots + e^{\ell_{V-1}}) - e^{\ell_1} e^{\ell_2}}{(e^{\ell_0} + e^{\ell_1} + \dots + e^{\ell_{V-1}})^2} \\ &= -\frac{e^{\ell_1}}{S} \cdot \frac{e^{\ell_2}}{S} = -p_1 p_2. \end{aligned} \quad (33)$$

c) Entries  $\frac{\partial p_2}{\partial \ell_1}$  and  $\frac{\partial p_2}{\partial \ell_2}$ : The same two computations with the indices swapped. For  $\frac{\partial p_2}{\partial \ell_1}$ :  $u = e^{\ell_2}$  is constant in  $\ell_1$  and  $\frac{\partial v}{\partial \ell_1} = e^{\ell_1}$  by (30); for  $\frac{\partial p_2}{\partial \ell_2}$  both numerator and denominator move, exactly as in (31):

$$\frac{\partial p_2}{\partial \ell_1} = \frac{0 \cdot S - e^{\ell_2} e^{\ell_1}}{S^2} = -p_2 p_1, \quad (34)$$

$$\frac{\partial p_2}{\partial \ell_2} = \frac{e^{\ell_2} S - e^{\ell_2} e^{\ell_2}}{S^2} = p_2 (1 - p_2). \quad (35)$$

d) *The pattern*: Arranged as a block of the Jacobian, the four entries are

$$\begin{pmatrix} \frac{\partial p_1}{\partial \ell_1} & \frac{\partial p_1}{\partial \ell_2} \\ \frac{\partial p_2}{\partial \ell_1} & \frac{\partial p_2}{\partial \ell_2} \end{pmatrix} = \begin{pmatrix} p_1 (1 - p_1) & -p_1 p_2 \\ -p_2 p_1 & p_2 (1 - p_2) \end{pmatrix}; \quad (36)$$

diagonal entries  $p_i (1 - p_i)$ , off-diagonal entries  $-p_i p_j$ . The general entry repeats the same quotient rule. Each  $v$ -derivative above is one instance of the general fact that exactly one term of the broken-out  $S$  involves  $\ell_j$ :

$$\frac{\partial S}{\partial \ell_j} = \frac{\partial}{\partial \ell_j} (e^{\ell_0} + \dots + e^{\ell_j} + \dots + e^{\ell_{V-1}}) = e^{\ell_j}. \quad (37)$$

The diagonal case  $j = i$  repeats (31) (the numerator also moves):

$$\begin{aligned} \frac{\partial p_i}{\partial \ell_i} &= \frac{e^{\ell_i} S - e^{\ell_i} e^{\ell_i}}{S^2} = \frac{e^{\ell_i}}{S} - \left( \frac{e^{\ell_i}}{S} \right)^2 \\ &= p_i - p_i^2 = p_i (1 - p_i). \end{aligned} \quad (38)$$

The off-diagonal case  $j \neq i$  repeats (33) (the numerator  $e^{\ell_i}$  is constant in  $\ell_j$ ):

$$\frac{\partial p_i}{\partial \ell_j} = \frac{0 \cdot S - e^{\ell_i} e^{\ell_j}}{S^2} = -p_i p_j. \quad (39)$$

**Result.** The two cases combine into one expression via the Kronecker delta, the standard notation for an entry of the identity matrix. It is the indicator (17) specialized to an index equality,  $\delta_{ij} = \mathbb{I}[i = j]$ :

$$\delta_{ij} = \begin{cases} 1 & i = j, \\ 0 & i \neq j, \end{cases} \quad (40)$$

which selects the diagonal case:

$$\frac{\partial p_i}{\partial \ell_j} = p_i (\delta_{ij} - p_j). \quad (41)$$

Check against (38) and (39):  $j = i$  gives  $p_i (1 - p_i)$ , and  $j \neq i$  gives  $-p_i p_j$ .

*B. Backward w.r.t. the logits*

**Derivation.** The upstream gradient  $\frac{\partial L}{\partial p}$  enters through the chain rule (1) summed over the row, one full column of the Jacobian (41) per output  $i$ . Broken out:

$$\begin{aligned} \frac{\partial L}{\partial \ell_i} &= \sum_{j < V} \frac{\partial L}{\partial p_j} \frac{\partial p_j}{\partial \ell_i} \\ &= \frac{\partial L}{\partial p_0} \frac{\partial p_0}{\partial \ell_i} + \frac{\partial L}{\partial p_1} \frac{\partial p_1}{\partial \ell_i} + \frac{\partial L}{\partial p_2} \frac{\partial p_2}{\partial \ell_i} \\ &\quad + \dots + \frac{\partial L}{\partial p_{V-1}} \frac{\partial p_{V-1}}{\partial \ell_i}. \end{aligned} \quad (42)$$

Substitute (38) and (39); the  $j = i$  term is the diagonal case, every other term is off-diagonal:

$$\begin{aligned} \frac{\partial L}{\partial \ell_i} &= \frac{\partial L}{\partial p_i} p_i (1 - p_i) + \sum_{j \neq i} \frac{\partial L}{\partial p_j} (-p_j p_i) \\ &= p_i \frac{\partial L}{\partial p_i} - p_i p_i \frac{\partial L}{\partial p_i} - p_i \sum_{j \neq i} \frac{\partial L}{\partial p_j} p_j. \end{aligned} \quad (43)$$

The middle term is exactly the missing  $j = i$  element of the sum, so it re-absorbs:

$$\frac{\partial L}{\partial \ell_i} = p_i \frac{\partial L}{\partial p_i} - p_i \sum_{j < V} \frac{\partial L}{\partial p_j} p_j, \quad (44)$$

where the remaining sum is the same one for every  $i$  in the row. Broken out in the same format, it is

$$\begin{aligned} \sum_{j < V} \frac{\partial L}{\partial p_j} p_j &= \frac{\partial L}{\partial p_0} p_0 + \frac{\partial L}{\partial p_1} p_1 \\ &\quad + \frac{\partial L}{\partial p_2} p_2 + \dots + \frac{\partial L}{\partial p_{V-1}} p_{V-1}. \end{aligned} \quad (45)$$

**Result.**

$$\frac{\partial L}{\partial \ell_i} = p_i \left( \frac{\partial L}{\partial p_i} - \sum_{j < V} \frac{\partial L}{\partial p_j} p_j \right), \quad i < V, \quad (46)$$

and  $\frac{\partial L}{\partial \ell_i} = 0$  for the padded region  $i \geq V$ . Summing (46) over the row, the leading  $p_i$  distributes into both terms; the

probabilities multiplying the (constant) dot product sum to one, leaving two copies of the same dot product:

$$\begin{aligned}\sum_{i < V} \frac{\partial L}{\partial \ell_i} &= \sum_{i < V} p_i \frac{\partial L}{\partial p_i} - \left( \sum_{i < V} p_i \right) \sum_{j < V} \frac{\partial L}{\partial p_j} p_j \\ &= \sum_{i < V} p_i \frac{\partial L}{\partial p_i} - \sum_{j < V} \frac{\partial L}{\partial p_j} p_j = 0.\end{aligned}\quad (47)$$

The gradient row always sums to zero, consistent with the shift invariance (26).

### C. Fused softmax + cross-entropy

The composition is the case that matters for the model: backprop through (18) and (25) together and watch the Jacobian collapse.

**Derivation.** Write out the chain per logit coordinate, full indices restored (upstream index  $i$  over the probabilities,  $j$  over the logits):

$$\frac{\partial L}{\partial \ell_{bt,j}} = \sum_{i < V} \frac{\partial L}{\partial p_{bt,i}} \frac{\partial p_{bt,i}}{\partial \ell_{bt,j}}. \quad (48)$$

Substitute what is already known into the combined equation: the cross-entropy gradient (24) for the first factor and the Jacobian (41) for the second:

$$\frac{\partial L}{\partial \ell_{bt,j}} = \sum_{i < V} \left( -\frac{\mathbb{I}[i = y_{bt}]}{p_{bt,i}} \right) p_{bt,i} (\delta_{ij} - p_{bt,j}) \quad (49)$$

The  $p_{bt,i}$  in the denominator cancels against the  $p_{bt,i}$  the Jacobian contributes:

$$= - \sum_{i < V} \mathbb{I}[i = y_{bt}] (\delta_{ij} - p_{bt,j}) \quad (50)$$

and the one-hot indicator keeps exactly one term of the sum,  $i = y_{bt}$ :

$$= -(\delta_{y_{bt},j} - p_{bt,j}) = p_{bt,j} - \mathbb{I}[j = y_{bt}]. \quad (51)$$

**Result.**

$$\frac{\partial L}{\partial \ell_{bt,j}} = p_{bt,j} - \mathbb{I}[j = y_{bt}], \quad j < V, \quad (52)$$

and  $\frac{\partial L}{\partial \ell_{bt,j}} = 0$  for the padded region  $j \geq V$ . The Jacobian collapsed entirely: no sum over the row survives in (52).

Cross-check via the general backward (46): with the cross-entropy upstream (24), the dot product evaluates to

$$\sum_{j < V} \frac{\partial L}{\partial p_{bt,j}} p_{bt,j} = \sum_{j < V} \left( -\frac{\mathbb{I}[j = y_{bt}]}{p_{bt,j}} \right) p_{bt,j} = -1, \quad (53)$$

a constant: for a one-hot target the row-wide weighted sum is known before reading a single element. Substituting  $-1$  for the dot product in (46), the  $p_{bt,i}$  outside the parentheses cancels the reciprocal inside them:

$$\begin{aligned}\frac{\partial L}{\partial \ell_{bt,i}} &= p_{bt,i} \left( \frac{\partial L}{\partial p_{bt,i}} - (-1) \right) \\ &= p_{bt,i} \left( -\frac{\mathbb{I}[i = y_{bt}]}{p_{bt,i}} + 1 \right) \\ &= p_{bt,i} - \mathbb{I}[i = y_{bt}],\end{aligned}\quad (54)$$

agreeing with (52): chaining through the Jacobian directly and substituting into the general backward are two routes to the same gradient. The zero-sum check of §IV also holds: the probabilities sum to one, and the indicator fires exactly once per row:

$$\sum_{j < V} (p_{bt,j} - \mathbb{I}[j = y_{bt}]) = 1 - 1 = 0. \quad (55)$$

The collapse did not need one-hotness. It needed only the standard-sum gradient (22) and the fact that the target is a distribution,  $\sum_{i < V} y_{bt,i} = 1$ . Repeating (48) with a general target distribution,

$$\begin{aligned}\frac{\partial L}{\partial \ell_{bt,j}} &= \sum_{i < V} \left( -\frac{y_{bt,i}}{p_{bt,i}} \right) p_{bt,i} (\delta_{ij} - p_{bt,j}) \\ &= -y_{bt,j} + p_{bt,j} \sum_{i < V} y_{bt,i} = p_{bt,j} - y_{bt,j},\end{aligned}\quad (56)$$

the one-hot result (52) being the special case  $y_{bt,j} = \mathbb{I}[j = y_{bt}]$ . Softmax + cross-entropy always backpropagates as *predicted minus target*.

### V. LINEAR LAYER (MATMUL)

Forward:  $Y = XW^\top + b$  with  $X \in \mathbb{R}^{(B \cdot T) \times C_{in}}$ ,  $W \in \mathbb{R}^{C_{out} \times C_{in}}$ ,  $b \in \mathbb{R}^{C_{out}}$ .

**Derivation. TODO:** Derive  $\frac{\partial L}{\partial X}$ ,  $\frac{\partial L}{\partial W}$ ,  $\frac{\partial L}{\partial b}$  from (1); check every shape. Note where the reduction over  $B \cdot T$  appears.

**Result. TODO:**  $\frac{\partial L}{\partial X} = ?$ ,  $\frac{\partial L}{\partial W} = ?$ ,  $\frac{\partial L}{\partial b} = ?$

### VI. GELU

Forward (tanh approximation, as in llm.c):

$$\text{GELU}(x) = \frac{1}{2} x \left( 1 + \tanh \left[ \sqrt{2/\pi} (x + 0.044715 x^3) \right] \right). \quad (57)$$

**Derivation. TODO:** Differentiate (57). Elementwise, so the chain rule is a pointwise product.

**Result. TODO:**  $\text{GELU}'(x) = ?$

### VII. LAYERNORM

Forward, per position  $x \in \mathbb{R}^C$  with scale  $\gamma$  and shift  $\beta$ :

$$\begin{aligned}\mu &= \frac{1}{C} \sum_i x_i, \quad \sigma^2 = \frac{1}{C} \sum_i (x_i - \mu)^2, \\ y_i &= \gamma_i \frac{x_i - \mu}{\sqrt{\sigma^2 + \varepsilon}} + \beta_i.\end{aligned}\quad (58)$$

**Derivation. TODO:** The interesting one:  $\mu$  and  $\sigma^2$  both depend on every  $x_j$ , so  $\frac{\partial y_i}{\partial x_j}$  has three terms. Work it via  $\hat{x}_i = (x_i - \mu)/\sqrt{\sigma^2 + \varepsilon}$ .

**Result. TODO:**  $\frac{\partial L}{\partial x_j} = ?$ ,  $\frac{\partial L}{\partial \gamma_i} = ?$ ,  $\frac{\partial L}{\partial \beta_i} = ?$

### VIII. ATTENTION

Forward, per head:  $A = \text{softmax}(QK^\top/\sqrt{h} + M)$ ,  $Y = AV$ , with causal mask  $M$  and  $Q = XW_Q$ , etc.

**Derivation. TODO:** Backprop in stages:  $\frac{\partial L}{\partial V}$  and  $\frac{\partial L}{\partial A}$  from  $Y = AV$ ; then  $\frac{\partial L}{\partial (QK^\top)}$  through the row-wise softmax (reuse §IV); then  $\frac{\partial L}{\partial Q}$ ,  $\frac{\partial L}{\partial K}$ . Track  $\sqrt{h}$  and the mask.

**Result. TODO:** Gradient for each of  $Q$ ,  $K$ ,  $V$ .

## IX. EMBEDDINGS (ENCODER)

Forward:  $x_{bt} = W_E[\text{tok}_{bt}] + W_P[t]$ .

**Derivation. TODO:** Gradients of a gather are a scatter-add; multiple  $(b, t)$  hitting the same token row accumulate. Connect to the  $+=$  convention of §I.

**Result. TODO:**  $\frac{\partial L}{\partial W_E} = ?$ ,  $\frac{\partial L}{\partial W_P} = ?$

## X. THE FULL BACKWARD PASS

Compose §III–§IX in reverse forward order, including the residual stream (where gradients add) and weight tying between  $W_E$  and the LM head.

**TODO:** Write the end-to-end gradient flow for one transformer block, then the whole model.

APPENDIX A  
GRADIENT CHECKING

Numerical check used to validate every derivation above:

$$\frac{\partial L}{\partial \theta_i} \approx \frac{L(\theta + \epsilon e_i) - L(\theta - \epsilon e_i)}{2\epsilon}. \quad (59)$$

**TODO:** Record per-dtype tolerances and how they relate to `tests/_dtypes.py`.

APPENDIX B  
USEFUL IDENTITIES

- $\frac{d}{dx} \log x = 1/x$ ;  $\frac{d}{dx} e^x = e^x$ ;  $\frac{d}{dx} \tanh x = 1 - \tanh^2 x$ .
- log-sum-exp and its gradient:  $\nabla \log \sum_j e^{x_j} = \text{softmax}(x)$ .
- **TODO:** Collect identities as they get used.